Circle the correct answer for each question. If you're unsure, you can write numbers 1, 2, 3, etc. next to the choices to get partial credit.

Question permutation table:

Version		1	2	3	4
Question					
1	→	9	1	2	1
2*	→	10	2	3	2
3	→	1	4	5	3
4	→	2	10	1	10
5	→	3	5	6	4
6*	→	4	6	7	5
7*	→	5	7	8	6
8*	→	6	8	9	7
9*	→	7	9	10	8
10	→	8	3	4	9

*: Question follows previous

The Women's Health Study included 39,876 female health professionals aged 45 years and older who were followed for an average of 10 years (Ridker et al., 2005). At the beginning of the study, participants were randomly assigned to take either a daily low-dose aspirin or a daily placebo pill for the duration of the study. At the end of the study, researchers measured the number of participants that suffered from a heart attack during the study.

Data are summarized in the following table.

	Low-dose Aspirin	Placebo	Total
Heart Attack	198	193	391
No Heart Attack	19,736	19,749	39,485
Total	19,934	19,942	39,876

1. Is a theory-based approach appropriate to evaluate the relationship between whether an individual took aspirin or a placebo and whether an individual suffers a heart attack?

   (a) No, since a simulation-based approach is always appropriate.
   (b) Yes, since 39,876 is larger than 20.
   (c) Yes, since 198, 193, 19,736, and 19,749 are all greater than 10.
   (d) Yes, since 391 and 39,485 are both greater than 10.

   Correct answers: (c)

   Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(a)	(d)	(c)	(d)
(b)	→	(d)	(c)	(b)	(a)
(c)	→	(c)	(a)	(a)	(b)
(d)	→	(b)	(b)	(d)	(c)

2. Based on a 95% confidence interval $(-0.0014, 0.0019)$ for the data above, what could be said for sure about the p-value for testing $H_0 : \pi_a - \pi_p = 0$ versus $H_A : \pi_a - \pi_p \neq 0$?

   (a) p-value < 0.01
   (b) p-value < 0.05
   (c) p-value < 0.10
   (d) p-value > 0.10

   Correct answers: (b)

Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(a)	(b)	(a)	(d)
(b)	→	(c)	(d)	(b)	(a)
(c)	→	(d)	(a)	(d)	(c)
(d)	→	(b)	(c)	(c)	(b)

3. For a confidence interval, if the confidence level from 95% to 99%, all else remaining the same, how would the center of the confidence interval be affected?

   (a) Increase
   (b) Remain the same
   (c) Decrease
   (d) Cannot be determined

Correct answers: (b)

Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(b)	(b)	(b)	(a)
(b)	→	(d)	(c)	(d)	(d)
(c)	→	(a)	(d)	(a)	(b)
(d)	→	(c)	(a)	(c)	(c)

4. For a confidence interval, if the confidence level from 95% to 99%, all else remaining the same, how would the the width of the confidence interval be affected?

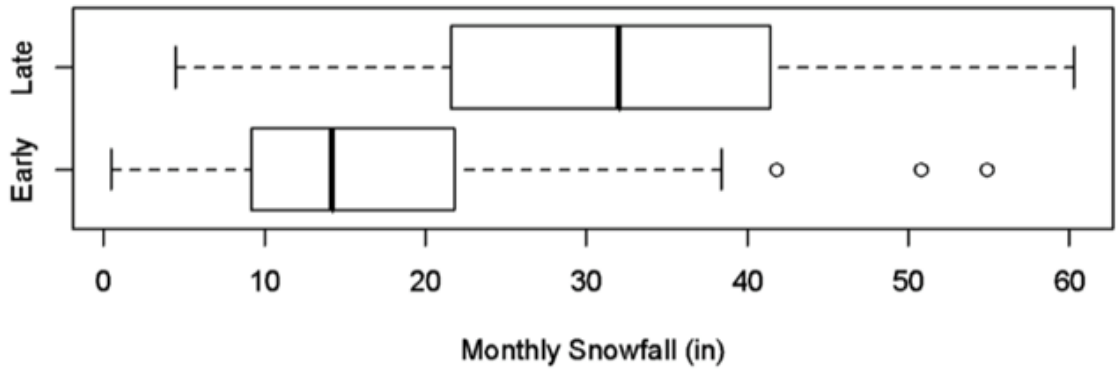
   (a) Increase
   (b) Remain the same
   (c) Decrease
   (d) Cannot be determined

Correct answers: (a)

Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(a)	(d)	(b)	(d)
(b)	→	(b)	(b)	(c)	(a)
(c)	→	(c)	(a)	(a)	(b)
(d)	→	(d)	(c)	(d)	(c)

Monthly snowfall (in inches) was measured over several winters in Fort Collins, Colorado. Researchers also recorded whether the measurement was taken in the Early winter (September to December) or Late winter (January to June). Boxplots displaying the distribution of monthly snowfall for each season are below.



Use this to answer the next five questions:

5. The shape of the distribution of monthly snowfall measurements for the Early season is

   (a) symmetric.
   (b) skewed right.
   (c) skewed left.
   (d) bimodal.

   Correct answers: (b)

   Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(a)	(c)	(a)
(b)	→	(d)	(c)	(d)	(d)
(c)	→	(c)	(b)	(b)	(c)
(d)	→	(b)	(d)	(a)	(b)

6. What is the upper quartile (Q3) of the distribution of monthly snowfall measurements for the Late season (approximately)?

   (a) 22
   (b) 33
   (c) 41
   (d) 60

   Correct answers: (c)

   Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(a)	(a)	(c)	(a)
(b)	→	(c)	(b)	(d)	(d)
(c)	→	(d)	(d)	(a)	(b)
(d)	→	(b)	(c)	(b)	(c)

7. Which of the following sentences correctly interprets the first quartile (Q1) of the distribution of monthly snowfall measurements for the Early season?

   (a) About 25% of Early season winter months in Fort Collins get less than 9 inches of snow.
   (b) About 50% of Early season winter months in Fort Collins get less than 9 inches of snow.
   (c) About 75% of Early season winter months in Fort Collins get less than 9 inches of snow.
   (d) About 25% of Early season winter months in Fort Collins get more than 38 inches of snow.

   Correct answers: (a)

   Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(c)	(a)	(d)	(c)
(b)	→	(d)	(c)	(a)	(b)
(c)	→	(b)	(d)	(c)	(d)
(d)	→	(a)	(b)	(b)	(a)

8. Which season has the larger median monthly snowfall?

   (a) Early
   (b) Late
   (c) The two medians are approximately equal.
   (d) The plot does not provide enough information to determine which median is larger.

   Correct answers: (b)

   Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(a)	(d)	(a)	(b)
(b)	→	(c)	(a)	(b)	(a)
(c)	→	(d)	(c)	(d)	(d)
(d)	→	(b)	(b)	(c)	(c)

9. The boxplots demonstrate that there is an association between which two variables?

   (a) Early and Late winter season
   (b) Snowfall and months
   (c) Monthly snowfall and whether the time was Early or Late winter season
   (d) Whether it was Early or Late winter season and months

   Correct answers: (c)

   Answer permutations: (All answers permuted)

Version		1	2	3	4
Answer					
(a)	→	(b)	(b)	(a)	(d)
(b)	→	(d)	(c)	(d)	(b)
(c)	→	(c)	(a)	(c)	(c)
(d)	→	(a)	(d)	(b)	(a)

10. Which of the following data sets has the largest standard deviation?

    (a) 1, 2, 3, 4, 5
    (b) 1, 3, 3, 3, 5
    (c) 1, 1, 3, 5, 5
    (d) 1, 1, 1, 1, 1

    Correct answers: (c)

Answer permutations: (All answers permuted)

	Version	1	2	3	4
Answer					
(a)	→	(c)	(a)	(c)	(d)
(b)	→	(b)	(c)	(d)	(b)
(c)	→	(d)	(b)	(b)	(a)
(d)	→	(a)	(d)	(a)	(c)